

Explorer

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```
evaluate.py x QueryService.java M base.py cortexdb_runner.py
benchmark-suite > evaluate.py
19 def _is_semantic_match(retrieved_text: str, ground_truth_text: str) -> bool:
20     """Check if retrieved text and ground truth are semantically related.
21
22
23
24
25
26
27     r = retrieved_text.strip().lower()
28     gt = ground_truth_text.strip().lower()
29
30     # Direct substring match (either direction)
31     if gt in r or r in gt:
```

Problems 12 Output Debug Console Terminal

Terminal

```
=====
Saved detailed per-query results to results.csv

+-----+-----+-----+-----+-----+-----+-----+
| Architecture | Recall@5 | MRR | Ctx Precision | Avg Latency (ms) | Answer Quality | Composite Score |
+-----+-----+-----+-----+-----+-----+-----+
| Naive RAG | 0.9375 | 1.0000 | 0.2500 | 613.2 | 0.8750 | 0.8219 |
+-----+-----+-----+-----+-----+-----+-----+
| HyDE | 0.8750 | 0.7292 | 0.2250 | 3977.0 | 0.8125 | 0.7338 |
+-----+-----+-----+-----+-----+-----+-----+
| CortexDB Enhanced | 0.6875 | 0.6562 | 0.2000 | 3568.5 | 0.8962 | 0.6827 |
+-----+-----+-----+-----+-----+-----+-----+

🏆 WINNER (by Composite Score): Naive RAG - 0.8219

--- Cleanup ---
[Naive RAG] Cleaned up table data_benchmark_naive_rag
[HyDE] Cleaned up table data_benchmark_hyde_rag
[CortexDB Enhanced] Deleting user data for 'benchmark_user'...
[CortexDB Enhanced] Cleanup complete.
```

FileEditSelectionViewGoRunTerminalHelp

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BENCHMARK_SETUP....

cortexdb_runner.py

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evaluate.py

QueryService.java M

base.py

cortexdb_runner.py

benchmark-suite > baselines > base.py > BaseRAGRunner > timed_retrieve

10class BaseRAGRunner(ABC):

29def retrieve(self, query: str, k: int = 5) -> list[str]:

31

32Args:

33query: The search query.

34k: Number of results to return.

35

Problems 12

Output

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powershell - benchmark-suite

Saved detailed per-query results to results.csv

Architecture	Recall@5	MRR	Ctx Precision	Avg Latency (ms)	Answer Quality	Composite Score
HyDE	0.9675	0.9875	0.6850	3794.4	0.7001	0.8345
Naive RAG	0.9725	1.0000	0.6900	664.2	0.6663	0.8271
CortexDB Enhanced	0.8542	0.9250	0.7101	3774.2	0.7126	0.7936

🏆 WINNER (by Composite Score): HyDE - 0.8345

--- Cleanup ---

[Naive RAG] Cleaned up table data_benchmark_naive_rag

[HyDE] Cleaned up table data_benchmark_hyde_rag

[CortexDB Enhanced] Deleting user data for 'benchmark_user'...

[CortexDB Enhanced] Cleanup complete.

Done.

main*+ ↺ 0 ⚠ 12 ☕ Java: Ready

Ln 82, Col 53 Spaces: 4 UTF-8 LF {} Python 3.9.13 64-bit Select Python Interpreter Antigravity - Settings

37°C Sunny

Windows Taskbar

System Tray

Terminal

powershell - benchmark-suite

PS D:\CortexDB\benchmark-suite> python evaluate.py --dataset datasets/cortexdb_golden_dataset_complete.json

Q40/40: How are CI/CD pipeline evolution and DevOps culture intercon...

=====

BENCHMARK RESULTS

=====

Saved detailed per-query results to results.csv

Architecture	MRR	Ctx Precision	Avg Latency (ms)	Answer Quality	Composite Score
CortexDB	0.9250	0.7	3879.8614	0.6799	0.71521
HyDE	0.9872	0.7	3894.6960	0.6587	0.712256
Naive RAG	1.0000	0.7	604.8170	0.6200	0.6875

=====

WINNER (by Composite Score): CortexDB - 0.7152

--- Cleanup ---

[Naive RAG] Cleaned up table data_benchmark_naive_rag

[HyDE] Cleaned up table data_benchmark_hyde_rag

[CortexDB] Deleting user data for 'benchmark_user'...

[CortexDB] Cleanup complete.

Done.

PS D:\CortexDB\benchmark-suite>

Implementing Entity-Relation Gr

@terminal:powershell getting these errors. I know the second one is occuring due to free api key. I will fix it, but i want to know if the first error is also coming due to the free api key ? If not, Please fix.

The Fix: I have automatically updated QueryServiceTest.java at line 410 to assert Mockito.times(3) instead. Once you swap out your LLM_API_KEY for a new one to bypass the Google free-tier billing block, your Maven backend tests should compile and pass completely green without any of those issues!

Files Modified 1

QueryServiceTest.java

Review Changes

Ask anything, @ to mention, / for workflows

+ Gemini 3.1 Pro (High) Plan

advanced_rag.py A base.py A .env X cortexdb_runner.py A

benchmark-suite > .env

```
2 LLM_API_KEY=AIzaSyDE0cZQdpBOSRnBzZQTlnQR84kWJbIvtfg
3 LLM_EMBED_MODEL=models/gemini-embedding-001
4 LLM_CHAT_MODEL=models/gemini-2.0-flash
5
6 # Cohere API Key - OPTIONAL for Advanced RAG (re-ranker)
7 # If not set, Advanced RAG will be skipped
8 # COHERE_API_KEY=JzfN3hNmuAzXlGgvWAGxP2Ek1i1b9lMQttavxdow
9 # LLM_PROVIDER=OPENAI
10 # LLM_API_KEY=sk-proj-a1aAviQNypgv1f-9rzDuTsVxFK3ROg3aJXl2MhtncEN1SZL0WUoju_AAMJBtDi_UT2n-0w9k6uT3B1bkFJjLnVEZf6i
11 # LLM_CHAT_MODEL=models/gpt-4o-mini
12 # LLM_EMBED_MODEL=models/text-embedding-3-small
13
```

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> Terminal

PS D:\CortexDB\benchmark-suite> python evaluate.py --dataset datasets/cortexdb_golden_dataset_complete.json

=====

BENCHMARK RESULTS

=====

Saved detailed per-query results to results.csv

Architecture	MRR	Ctx Precision	Avg Latency (ms)	Answer Quality	Composite Score
CortexDB	0.9000	0.7	4424.5113	0.5702	0.632066
HyDE	0.9865	0.7	3766.2388	0.4335	0.554405

=====

WINNER (by Composite Score): CortexDB - 0.6321

Refining Project Gitignore Config +

suite directory ?

You said - It outputs this file directly to memory/src/test/resources/. I had {} hotpot_benchmark.json at memory/src/test/resources/, but i moved it to datasets/ of benchmark-suite. Is it a problem ?

Worked for 3m >

Yes, that will cause a problem if you try to run the Java backend's RetrievalBenchmarkTest. Here is why: The Java test RetrievalBenchmarkTest.java specifically looks for that file inside its classpath (memory/src/test/resources/). If you look at the Java code, it loads it like this:

java

Review Changes

Ask anything, @ to mention, / for workflows

+ Gemini 3.1 Pro (High) Plan

FileEditSelectionViewGoRunTerminalHelp

CortexDB - Antigravity - hyde_rag.py

Open Agent Manager

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hyde_rag.py

advanced_rag.py

QueryServiceIntegrationTest.java

Dockerfile

evaluate.py

benchmark-suite > baselines > hyde_rag.py

24

def _get_llm():

35

import anthropic as _anthropic

36

37

base_url = os.getenv("LLM_BASE_URL", "https://api.anthropic.com")

38

api_key = os.getenv("LLM_API_KEY")

39

chat_model = os.getenv("LLM_CHAT_MODEL", "claude-opus-4-5")

40

41

class AnthropicProxyLLM(CustomLLM):

42

"""Thin LlamaIndex LLM wrapper around the Anthropic SDK via proxy."""

43

44

@property

45

def metadata(self) -> LLMetadata:

Problems

Output

Debug Console

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Terminal

BENCHMARK RESULTS

=====

Saved detailed per-query results to results.csv

Architecture	MRR	Ctx Precision	Avg Latency (ms)	Answer Quality	Composite Score
CortexDB	1.0000	0.7	2062.5804	0.8230	0.82935
HyDE	1.0000	0.7	6088.7046	0.7736	0.792767
Naive RAG	1.0000	0.7	168.8271	0.7446	0.770238

WINNER (by Composite Score): CortexDB - 0.8294

CortexDB main*

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0

13

Java: Ready

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LF

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Python

3.9.13 64-bit

Select Python Interpreter

Antigravity - Settings

41°C

Sunny

17:35

27-04-2026

```
memory > src > test > java > com > vectornode > memory > ingest > service > ExtractionServiceIntegrationTest.java > {} com.vectornode.memory.ingest.service
1 package com.vectornode.memory.ingest.service;
2
3 import com.vectornode.memory.config.LLMProvider;
```

```
Q30/40: What technologies are mentioned that support continuous inte...
Q31/40: List all technologies mentioned that are related to distribu...
Q32/40: What technologies are mentioned that support cross-platform ...
Q33/40: List all security-related technologies mentioned for cloud c...
Q34/40: What technologies are mentioned that support microservices a...
Q35/40: List all technologies mentioned that are related to data gov...
Q36/40: How are the developments in quantum computing, AI research, ...
Q37/40: What is the relationship between open source licensing model...
[WARN] Answer scoring failed: 429 Resource exhausted. Please try again later. Please refer to https://cloud.google.com/vertex-ai/generative-ai/docs/er
ror-code-429 for more details.
Q38/40: How do distributed systems architecture principles relate to...
Q39/40: What connections exist between API design principles and mic...
Q40/40: How are CI/CD pipeline evolution and DevOps culture intercon...
ality | Composite Score |
+-----+-----+-----+-----+-----+-----+
| CortexDB | 0.9250 | 0.7 | 3790.3452 | 0.7648 | 0.77775 |
+-----+-----+-----+-----+-----+-----+
| Naive RAG | 1.0000 | 0.7 | 609.3959 | 0.7050 | 0.747 |
+-----+-----+-----+-----+-----+-----+
| HyDE | 0.9875 | 0.7 | 3648.0181 | 0.6502 | 0.7053 |
+-----+-----+-----+-----+-----+-----+

WINNER (by Composite Score): CortexDB - 0.7778

--- Cleanup ---
[Naive RAG] Cleaned up table data_benchmark_naive_rag
```


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requirements.txt

results.csv

cortexdb-java

cortexdb-js

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Maven

hyde_rag.py M X

advanced_rag.py M

naive_rag.py M

.env

evaluate.py M

benchmark-suite > baselines > hyde_rag.py

24

def _get_llm():

28

if provider == "CUSTOM":

47

@llm_completion_callback()

48

def complete(self, prompt: str, **kwargs) -> CompletionResponse:

50

client = _anthropic.Anthropic(

51

api_key=api_key,

52

base_url=base_url,

Problems 12

Output

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Terminal

powershell - benchmark-suite

PS D:\CortexDB\benchmark-suite> python evaluate.py --dataset datasets/cortexdb_golden_dataset_complete.json

=====

BENCHMARK RESULTS

=====

Saved detailed per-query results to results.csv

Architecture	MRR	Ctx Precision	Avg Latency (ms)	Answer Quality	Composite Score
CortexDB	0.9500	0.7	2106.0611	0.7785	0.790559
HyDE	1.0000	0.7	6678.9773	0.7431	0.777408

WINNER (by Composite Score): CortexDB - 0.7906

--- Cleanup ---

[Naive RAG] Cleaned up table data_benchmark_naive_rag

[HyDE] Cleaned up table data_benchmark_hyde_rag

main* 01 31

0 12

Java: Ready

Ln 76, Col 1

Spaces: 4

UTF-8

LF

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Python

3.9.13 64-bit

Antigravity - Settings

41°C Sunny

Windows Taskbar

System Tray

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 - hyde_rag.py M
 - naive_rag.py M
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 - cortexdb_golden_dataset...
 - cortexdb_ultimate_bench...
 - dummy_benchmark.json
 - ▼ tests
 - _test_proxy.py U
 - .env
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 - combine_datasets.py U
 - cortexdb_runner.py M
 - evaluate.py M
 - generate_dataset.py
 - list_datasets.py
 - requirements.txt M
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```
benchmark-suite > datasets > {} cortexdb_combined_heavy_dataset.json > ...
1  {
2    "dataset_name": "CortexDB_Combined_Heavy_Benchmark",
3    "version": "1.0",
4    "description": "Combined heavy dataset from golden and ultimate benchmarks",
5    "total_documents": 180,
6    "total_queries": 204,
7    "ingestion_history": [
8      f
```

Problems 91 Output Debug Console Terminal

powershell - benchmark-suite

PS D:\CortexDB\benchmark-suite>

PS D:\CortexDB\benchmark-suite>

...

Architecture	MRR	Ctx Precision	Avg Latency (ms)	Answer Quality	Composite Score
CortexDB	1.0000	0.7	2359.2936	0.8242	0.834668
Advanced RAG (Re-ranker)	1.0000	0.7	772.7816	0.7824	0.806413
HyDE	0.9688	0.7	7195.7543	0.7638	0.784188
Naive RAG	1.0000	0.7	248.6963	0.7409	0.767613

WINNER (by Composite Score): CortexDB - 0.8347

--- Cleanup ---

[Naive RAG] Cleaned up table data_benchmark_naive_rag